



Fix the Order

Quick facilitation notes & answer key

Goal

Children test a buggy arrow program, find the wrong/missing/extra step, and fix it so Bit reaches the star. Testing and correcting a sequence is debugging — the K-appropriate close to the sequencing arc.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children detect and correct an error in a sequence of steps.

No reading required of the child: read each prompt aloud; the child circles/crosses out and draws the fix.

Materials

The printed worksheet + a pencil. Optional: floor boxes to walk the buggy program and feel where it goes wrong.

Run it (about 15 min)

1. MODEL: walk → ← → from box 1 — box 2, box 1, box 2. It ends on box 2, not the star. The ← is the bug; fix to → →.
2. GUIDED: Exercise 1 — the program is one step short; add a → so it is → → → to box 4.
3. YOU TRY: remove the extra arrow (Ex 2 → →), then fix the wrong first arrow (Ex 3 → → →).
4. Connect: 'When a plan does not work, find the wrong step and fix it — that is debugging.'

Answer key (modelled by the run-gate)

Model — → ← → lands box 2; the ← is wrong; fix to → → (box 3).

Ex 1 — → → reaches only box 3; a step is missing; fix to → → → (box 4).

Ex 2 — → → overshoots to box 3; remove one arrow; fix to → (box 2).

Ex 3 — ← → → lands box 2; the first ← is wrong; fix to → → (box 3).

Watch for & differentiate

Common slip: changing a correct step — have them walk the program first to see exactly where it fails.

Simplify: use only 'missing/extra' bugs before the wrong-direction bug.

Extend: have a child write a buggy program for a partner to debug.

Movement option: walk the buggy steps as a class and stop the instant Bit is off track.

Success looks like

The child identifies the wrong/missing/extra step and draws a corrected program that lands Bit on the star.